receiver: reading-group dependencies
Rows depend on columns. U = uses; A = assumes; UA = both.
Numbers match PDF contents; E = external module.
Blank: no cross-group edge. Grey: internal or external detail omitted.

Group / prerequisite column	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16	17	18	19	20	21	22	E1
1 Modes, Fock space and the mode algebra	-																						
2 The receiver instrument and its physical model	U	-									A												
3 Component graphs and compilation	U	U	-								A												
4 Port Grams and the click-pattern POVM	U	U		-							A												
5 Representation independence and coarse-graining	U	U		U	-																		
6 The analytic high-photon bound	U			U		-		U															
7 Exact block eigenvalues	U			U		U	-																
8 Setting families and the emitted weight	U	U		U		U	U	-			A												
9 The reference, enclosed and exact	U	U	U	U	U				-	U													U
10 Residuals, tolerances and floating-point evaluation		U	U	U	U	U	U	U	U	-	A												U
11 Assumptions	U						U	U		U	-												U
12 Requirements: compilation, representation and outcomes		U	U	U	U							-											
13 Requirements: the emitted weight						U	U	U	U	U			-	U									U
14 Requirements: residuals, tolerances and the evaluation label		U	U	U		U		U	U	U			U	-									U
15 Requirements: the assumptions													U		-								
16 Component catalogue: Catalogue entries and their admissibility	U	U	U						U	U	A					-							
17 Component catalogue: A worked BB84 receiver	U	U	U	U	U	U	U	U								U	-						
18 Detector imperfections: Detector efficiency	U		U	U	U					U	A							-					
19 Detector imperfections: Dark counts	U			U		U		U		U	A								-				
20 Detector imperfections: The dark-composed family and the noise floor	U	U		U	U	U		U	U									U	U	-			U
21 Detector imperfections: Efficiency normalization	U	U	U	U	U			U			A							U	U	U	-		
22 Detector imperfections: Requirements									U	U				U				U	U	U	U	-	U
E1 foundations (external)	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-	-